ENVIRONING MEDIA

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ENVIRONING TIME

Mediating climate modeling

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Introduction

Is the Anthropocene about the past or the future? Being interested in the temporalities of climate change, I sometimes wonder. The age of humankind should arguably be about both, not only capturing the geological era when humans made a permanent impact on Earth, but also incorporating a future, where we struggle to manage this impact. Often enough, however, I get the impression that the Anthropocene is mostly about the past and where to put the spike in the horizontally organized geological calendar of the Earth, the strata of previous times which display the imprint from earlier flora and fauna, used by scientists to make order in change and create chronologies. Western intellectuals have indeed engaged with the long term since geologists started to question the Biblical description of time,[1] long before the term Anthropocene. When physicists formulated the idea of the Big Bang, time also took center stage. Recent writings that address this extreme long term have reached a rather broad audience.[2] Ironically, or perhaps congenially, this historical interest coincides with our inability to manage climate change. It is as if we find comfort in the fact that we did have a history, even if said change has made our future hazy.

This historic focus is of course legitimate and important. The tension between the age of humans and the extreme long term of the geological gaze holds promise for the intellectual work we are facing as planet-altering creatures. To place ourselves alongside dinosaurs are not only attractive for children in a particular age, but it also affords a more holistic perspective of stunning complexity. If we could envision time and understand change in temporal perspectives of varying scales, much would be gained for humankind. However, the future is also important for us to envision and understand. And there is an ongoing battle for the

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future, played out by different actors in different genres, resulting in a multitude of possible futures for humanity.[3]

I am interested in this future – or rather these futures – and how time and possible futures are mediated through scientific practices. I suggest that we can regard these practices as *environing technologies*, that is, technologies that create environments. Environment is normally understood as a spatial concept, but I argue that it is also temporal, working with time and change. The chapter will discuss how we can understand these futures brought to us by one of the most prominent organizations in the business, namely the Intergovernmental Panel on Climate Change. There are indeed many ways to talk about the future, but for climate change, *models* have become central. Models are key epistemic tools in climate science and the basis for much of the politics surrounding climate-change action and mitigation.[4] The model results are often mediated through classic *visual techniques* of the natural sciences, such as graphs and diagrams.[5] In this chapter I will look closer at how these visual techniques work and what they might convey regarding the temporalization of climate change. This temporalization fundamentally affects our perception of the planet in the years to come, a perception which is key in the response to climate change and should therefore be of central importance. In the light of their importance, I contend that we have not paid enough attention to these temporalities.

In the following I will first discuss how some of the results from the IPCC assessments are presented visually in the reports, and how we can understand the temporalities in play. I will then present the framework of environing technologies and its possible usefulness, and finally, I will discuss if environing media could be an alternative concept when analyzing the graphs that convey the output from climate-change models.

Modeling and mediating the future

Climate-change models frame the climate issue for humankind and structure our appreciation of what is to come. The futures that these models allow enable an analysis of the temporal dimensions of climate change.[6] The main actor is the Intergovernmental Panel on Climate Change which was created as a sub-organization to the United Nations in 1988. The IPCC releases assessments with some regularity, putting together state-of-the-art knowledge on climate change which will guide the international system in policy decisions. The First Assessment Report was published in 1990, the fifth in 2014.[7]

The IPCC reports have undergone change since the first publication in 1990. At first, they were graphically simple without color and relatively sparse with illustrations other than graphs and tables. In the Second Assessment Report (1995) blue was introduced and in the Third Assessment Report (2001) full color was used, which allowed for more elaborate illustrations and visualizations. There is no clear tendency in the number and type of graphic aids other than that complexity increases over time and that more dimensions are incorporated

into a common frame. The publication of the report has increasingly attracted attention of news media and the Summaries for Policy Makers (SPMs) most likely have a much larger readership than the full scientific reports. The images from the SPMs therefore enjoy a greater circulation in the public sphere and are thus a focus of my interest.[8]

Climate change and its consequences are increasingly difficult to grasp. It is not primarily the sheer science of the change that is challenging, which in earnest is not too complicated regarding the basic processes, but the scale of change and the intricate relations between our ordinary lives and the consequences for people on the other side of the globe. Individual and collective emissions in the small scale together and over the long term create an aggregate impact which transcends the planetary boundaries and remains forceful over many years to come.[9]

The mediation of climate change has a long history and is deeply politicized in many parts of the world.[10] In particular, news media has played a central role in conveying the scientific knowledge on climate change, but arguably failed on many arenas as the main message often was one of disunity among scientists when there in fact have been an almost total agreement on the main points for a very long time.[11] Even when true to scientific consensus it has proven difficult for the issue of climate change to take center stage as the crisis it is.

Time in relation to climate-change science has in general attracted less attention in the reflective and critical scholarship on climate change, where uncertainty, inertia, and denial have been thoroughly studied.[12] This is somewhat paradoxical as time is central to change. We know that the climate is changing because we can compare with earlier climates, and our survival depends on our ability to manage the change to come. Still, the temporal dimension is shadowed by sea-level rise, global mean temperatures, and CO_2 concentrations in the atmosphere – all three elusive enough, yet more concrete than time.

The greenhouse effect, and later climate change, was for long portrayed and understood as something that would happen in the distant future. In fact, this view was also held by scientists who based their decisions on target years allowing model results to be noticeable. Central to much early climate-change science was to establish if and how much of the change was anthropogenic and what could be attributed to natural variation, i.e. fluctuations that depend on astronomical, geological, chemical, and other processes that are not connected to human existence. Since climate change is a slow process, the consequences would indeed lie in the future. Thus, when modeling started in the 1960s and 1970s, a certain timeframe was required to get a significant signal. In the First Assessment Report of the IPCC, which was published in 1990, this was framed as a need: "100 years or more are necessary to support study of potential anthropogenic impacts on the climate system".[13] A target year far in the future was hence central.

The choice to model the long term was dependent not only on the needs for a significant signal but also on the tools available. Computational power increased exponentially from the 1960s and onward, i.e. doubling regularly, allowing for a greater number of components and variables to be included in the simulations.

A great challenge early on was the poor spatial resolution, due precisely to computer capacity, which resulted in very crude estimates and averages. Increased spatial resolution contributed to more useful results. Extremely important was to also incorporate more complex processes simulating the climate system. Early models were so-called equilibrium models that calculated how the climate system reacted on a doubling of CO_2 concentration in the atmosphere. Later, so-called transient models would instead work with constantly changing CO_2 concentration that would better reflect the changes in the system.[14]

In the First Assessment Report, the dominant target year of the scenarios building on climate-change models was 2100. In fact, in the following assessments up until the Fifth Assessment Report, published in 2014, the target year of 2100 has been the most common one.[15] There are reports in which also the year 2300 is featuring, like the Third Assessment Report, but in comparison to the number of figures where the x-axis ends on 2100, they are very few.

I propose that putting 2100 as target year for the scenarios has consequences for how we understand the futures humankind is facing. It frames the issue temporarily. The unintended paradox of choosing 2100, a year that would allow a large enough signal, is that it is so distant in the future that none of those who conceived of it will be alive then. This is also true for the politicians and business leaders who make decisions today. Thus, a year that works for science does not work for policy. To some extent, we can regard this as a matter of scale. Anne Pasek has framed this as a tug of war between "scales germane to the problem and scales more appropriate to mobilizing political action".[16] Depending on how we articulate the problem at hand, the mediated message becomes elusive. Let me elaborate further in the next section.

The power of the graph: 2100 as the end of time

Lynda Walsh has taken an interest in the visual rhetoric of climate change and proposes that graphs make an argument that supports a scientific claim.[17] When it comes to modeled change over a chosen timeframe, we must ask what the scientific claim of such a graph consists of. In the graphs in focus here, the particular claims vary, since they deal with different change in temperature, sea-level rise, ppm, etc. What unites them is how far into the future the models run. Thus, the repeated claim is arguably that 2100 is the most important temporal frame of climate change.

James Risbey has criticized this focus and called it a "2100ism", which has created a paradigm for the climate-change issue with severe consequences. Risbey argues that this focus "obscures the connection between impacts beyond 2100 and policy actions in the present".[18] Since impact will continue well into the coming centuries, stopping at 2100 gives us a wrongful picture of the futures at hand and might even lead to an unwillingness to act now. At the same time, Risbey acknowledges that 2100 is convenient as it connects to human timescales.[19] The year 2100 serves some functions well, and other poorly.

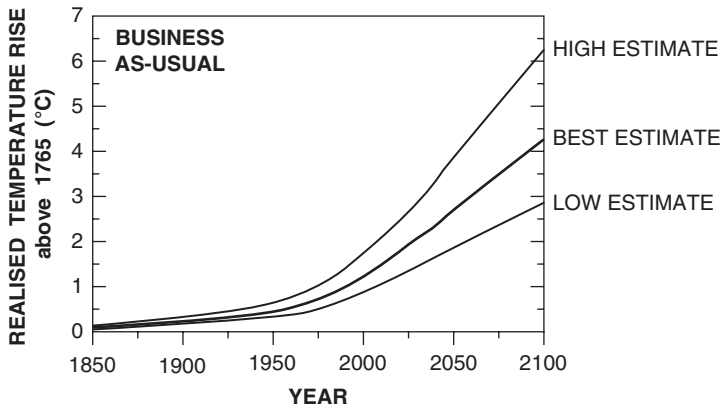

FIGURE 8.1 First Assessment Report, *Policymaker Summary of Working Group I, section 5.1, page 74. IPCC 1990.*

I would argue, however, that even though this target year is far away in the future, it can also be seen as the time *when* climate change happens. Indeed, that is by no means what the graph tells us. However, such a misunderstanding might not be as farfetched as one might think. Let us look at a graph from the First Assessment Report (Figure 8.1).

In the graph from the First Assessment Report, three business-as-usual scenarios are displayed, and we can clearly see that the best estimate will result in a temperature rise above 4 degrees in 2100. The lines leading up to the three resulting temperatures can be used to trace changes earlier in time, such as 2050 for example, but it is difficult and unclear – one would almost have to use a ruler.

In her elaboration on the so-called hockey-stick curve, the temperature change from the year 1400 until the present, Anne Pasek has talked about *charismatic data*.[20] The graph, building on a plethora of sources, displayed very forcefully the abrupt change in average temperatures on the Northern Hemisphere and had a wide circulation far beyond the first publication in *Nature* in 1998 and the following prominent role in both the Third Assessment Report and Al Gore's film "An inconvenient truth". The graph became iconic and managed to convey complicated scientific results in an image that could not be misunderstood.[21]

The examples of Pasek deal with data from the field rather than modeled data. However, *I argue that also modeled data can be charismatic and convey messages beyond the mere figure*. In fact, these graphs often do not make a clear distinction between what is modeled and what is measured. The IPCC reports frequently merge historically measured and re-analyzed data with model projections resulting in graphs that stretch both backward and forward from today. In the First Assessment Report the graph above displayed temperatures from 1850 to 2100 depending on various scenarios.

Building on Walsh and Pasek, I suggest that the target year attracts attention through its position in the graph, where it displays the end of time and what will

happen at that point. As mentioned, in the following assessment reports, the visual arguments become more complicated. In the Fifth Assessment Report, color is widely used, and the graphics are elaborate and complex. Graphs are combined with bars and maps and include a geographic overview of the impacts on a continental level. But the bottom line is more often than not the same. Several scenarios are displayed and result in temperature change that is projected up until 2100.

The fact that the target year has stayed the same since the first report in 1990 is striking and remarkable, and furthermore adds to the message that this is what we must deal with. There are obvious pros of maintaining the same target year as comparison becomes easy, but at the same time, keeping the target year means that we approach the end. Side by side and one after another, these reports covey a steady message: the future of humankind is shrinking.

This was not what modelers imagined when choosing 2100 as target year. However, some did recognize that model time could be connected to culture time and thereby different temporalities synchronized, to use a concept by Helge Jordheim.[22] The year 2100 ends a century and in that capacity carries historic meaning. The endings of centuries or even millennia are points in time where stories abound about change being ushered in and about the old giving way to the new. They translate into a human scale of generations and forefathers. It is possible to remember the turn of the century as something told by those before us, or something experienced as a transition by ourselves. The year 2100 indeed frames the climate issue in a temporal context that is possible to grasp for humans.

At the same time, it is noteworthy how seldom this year features in the public discourse on climate change. It totally dominates the assessment reports but only rarely becomes the focus of a daily discussion in the public sphere. Instead, other temporalities have become more functional, other scales more apt. The Sustainable Development Goals, for example, have 2030 as their target and the period 2020–2030 has been declared a "Decade of Action". This is a timeframe easy to overview and allowing for different goals to be broken down.[23] Yet another example is the Paris Agreement, in which the signatories to the Convention strive to keep warming below 2 degrees, and ideally 1.5 degrees. This shifts the focus from scenarios for 2100 to the effects of the absolute and total emissions of greenhouse gases and instead results in a global carbon budget. This in turn translates the climate issue to a so-called stock problem, giving rise to an understanding of time as running out but in a much shorter timeframe than 2100 given the present emission rate. We will soon use up our budget long before 2100 and the end is brought closer.[24] In relation to the Paris Agreement, a 2100-ism has been replaced by a deadline-ism.

Environment, technology and media: What is in a concept?

Let me now move to a discussion of concepts and of environing. A central starting point for this chapter is the claim that environment is a human creation, whereas nature is not. The process of creating environment can be captured by

the verb environ, an early modern word describing how land was encirculated and domesticated.[25] Etymologically, the word has siblings in many roman languages. Environment is the result of environing, and the action requires technology. To stake out your territory, enclose the sheep, and screen off the fox, you need the technology of the fence. Human settlement in various forms, farming and agriculture, mobility, and transport – all need technology that shape and structure nature into environments. Over time and with changing scale environing technologies have come to include enormous and complex tools, networks, and artifacts.[26] They include gigantic mines above and below Earth, oceans filled with vessels, wrecks and plastic, a space occupied by satellites and debris. The Anthropocene is the ultimate illustration of the planet as environment. Our alterations have become global and impact not just the entire Earth, but also reach into outer space and the bottom of the sea.

When we assess the environmental impact of technology, terraforming tools are the most common of technologies that environ. These environing technologies *shape* and re-shape our physical world, sometimes turning the planet inside-out as Gabrielle Hecht has put it.[27] They involve devices that allow us to build not just the shaky fence but also the Chinese wall, the dirt road and the autostrada, to cultivate not just the garden but to manage the industrialized agriculture of the 21st century. Indeed, the Anthropocene Garden could be understood as a monumental scaling of the original enclosure of the land around a settlement.[28]

There are, however, also environing technologies that work with conceptualizations and framings. The *shaping* environing technologies mentioned above can be supplemented with environing technologies that *sense* and *write*. These two terms should be understood broadly as collecting and conveying, or with key concepts in the information age: as input and output. Often these different kinds of technologies are connected and work together to form a resulting environment, which is both materially and conceptually different. The Anthropocene is in itself a conceptualization of the Earth, not just the result of shaping practices of the Holocene. As such, it might be just as powerful in helping us understand the human condition in relation to the material world. The word itself does environing work as it allows us to perceive and grasp these alterations. However, even if the shaping dimensions are limited, sensing and writing are often – but not always – part and parcel of the same environing. To be able to write down sustainable development goals, a sensing of the dimensions and the possible targets is a prerequisite. To paint a picture that claims to resemble the archipelago, you must have seen it or had someone tell you about it.

There are many such environing technologies that work primarily with our appreciation and understanding rather than with materially transforming and shaping. These conceptual environing technologies can also be epistemic, economic, existential, and emotional. Ecosystem services, for example, is a conceptualization of the environment that takes economy as baseline, translating the value of species, habitats, and a raspberry thicket on a clear-felled slope to something countable, quantitative, possible to factor into plans and politics. Red lists

are not just written estimations of species in danger, but epistemic objects that highlight certain parts of flora and fauna and thereby influence our understanding of biodiversity loss and change. Moreover, they show us global degradation and they can be deeply emotional as well as existential. Many writing environing technologies have the potential to work with our emotions through the different available genres. Film, literature, and art can all speak to us on levels that are not primarily intellectual.

In the previous sections I have focused on certain types of *writing* environing technologies that mediate an understanding of climate change, namely graphs. Is it possible to call these writing environing technologies for environing media? And what would be gained with such a term? To answer that we need to look closer at media and technology and their respective students.

Media and mediation are non-trivial concepts. The saturation of media in the present society has been termed mediatization, to separate it from the more specific term mediation which refers to the process of conveying.[29] That mediation and mediatization demand various technologies and practices of different kinds adds to the complexity of any analysis. Moreover, media are not just mass media, but involve forms of interpersonal and group communication as well as many possible hybrid forms. These forms differ within themselves in turn also because of the technologies involved. Susan Douglas argues that the relation between media and technology has been differently understood by media scholars and theorists on the one hand and historians and sociologists of technology on the other, and she contends that this has been the case for over half a century. Seemingly determinist stances from a global and macro-perspective have been challenged by micro—case studies deeply influenced by a constructivist approach.[30] Ian Hutchby has suggested the term *affordances* which allows technology to enable and limit without arguing that it totally determines the outcome, escaping the detested theory of technological determinism.[31]

The usefulness of a theoretical concept rests on its ability to capture and explicate something that through this concept becomes known in a new way. However, whether it catches on in a discipline or even manages to transcend different fields of knowledge also depends on how it resonates with previous and ongoing scholarly debates and approaches. That affordances have gotten greater traction within history and sociology of technology is much due to the discussion on *what technology does* in the scholarly work of those disciplines. Within the much larger scholarly domains of media and communication studies, *what media does* is similarly central. The concepts *technology* and *media* are in themselves and respectively foci of debates and discussions in the respective fields; or put differently, humanist and social science scholars interested in technology and media have both broad and deep understanding of the central concepts of their interests. The opposite is likely less true.

This could mean that *writing environing technologies* and *environing media* surface as two very different conceptual constructs in certain domains, whereas they are almost exchangeable in others. Depending on what you put into media and

technology respectively, and where you place your discussion, the understanding and analytical power might shift. Whereas an extensive and deep discussion of the meaning of technology is limited to a few narrow disciplinary fields, clearly the media concept has recently been altogether challenged by an expansion that transcends earlier understandings even outside of media and communication studies. The scholarship on elemental media by John Durham Peters is a useful example.[32] The coining of the term environing media by Johan Gärdebo and Adam Wickberg builds partly on this broader understanding of the term media. See further the introduction to this volume.

The consequences of the fact that technology and media are contextual might be that an innovative use of media in connection to environing could have greater traction than the corresponding compound writing environing technologies, both because media studies engage more academics and students than do technology studies and because the media concept in itself is already undergoing change. I suggest, however, that writing environing technologies is a broader concept than environing media, even when media is understood in an expansive way. This is because of the reach of the term technology, which can incorporate much of human activity and practice. Still, in certain contexts environing media might do precisely the work needed from a concept, which is to capture and explicate and allow for an understanding of something known in a new way. In the next section I will discuss whether the concept of environing media can be used to analyze how time, futures, and change are conveyed through scientific practices of mediation and how that in turn environs. I will use the term media and mediation in a more traditional and limited way than the recent one discussed above and in the introduction to this volume. My aim is to see if environing media can do a different job than writing environing technologies and if so, what the results might be.

Graphs as environing media?

I have argued that the graphs in the IPCC summary for policy makers frame the climate issue temporally. These graphs in turn build on results from the climate models and scenarios which are central in climate-change science. Can the concept of environing media be usefully employed to understand the work that these practices carry out? As detailed above, conceptual environing technologies work with our understanding of the environment, creating and framing it in a particular way that allows for certain actions and limits other. The examples that have previously been put forward, however, all work with the physical and spatial environment, and even though change is always temporal, time in itself has not been in focus for such an analysis by scholars interested in environing. A key issue is if we would accept time as a dimension that forms environment – in other words, something that with the help of technology can be environed.

If we regard the Anthropocene a term that is a writing environing technology, and we also accept that the Anthropocene is a temporal concept, it will

mean that environing could also be temporal. The interest in history that is now growing illustrates this, and puts human activity in perspective, challenging, in fact, what it is to be human. This interest is increasingly matched by parallel interest in the futures of humanity, of which the most spectacular is perhaps the plans of re-settlement on Mars, a plan B nourished both by a specific idea of progress and the understanding that Earth one day will have to be abandoned. These plans also work with timeframes that are distant and often unprecise, albeit much shorter than those for history. In both cases, time structures the understandings and possibilities of humanity, central to environing. Thus, I would suggest that environment is not just spatial but also temporal and that there are technologies that also work temporally, environing human existence, past and future. This is just as important as any physical environing.[33]

Climate models are complex technologies with encompassing input, intricate computational labor, and advanced output, which can be thought of as conceptual environing technologies. The models and scenarios themselves do not mediate or transform in any straightforward way; this is done by visual representations of the results from models and scenarios. In the form of graphs, temporalities that are of interest for human action in the Anthropocene are displayed. The graphs *mediate* the numerical results of the model runs and the scenario work and could therefore be regarded as *environing media* rather than environing technologies. This would be in line with a more general understanding of the visual as media in climate-change communication. This would also be true for the bucket, which transforms the stock problem to a flow problem, and mediates the proposal that time is running out. One could argue that the entire IPCC reports are environing technologies as they indeed frame the climate issue. The temporal environing, however, is specifically achieved by the visual mediation.

I suggest that the term environing media can be used to capture the temporal work that the IPCC graphs do. The term might furthermore be more attractive than environing technology since the mediation is a key function of the graph, a particular visual media technology. Environing media in this case underlines that there is ongoing translational work done by the graph, which is not conveyed by the term writing environing technology. Even though it is possible to further detail the technologies involved by introducing the word communication, to stack terms in that way is less attractive. Using the term environing media makes the process more specific and precise.

Conclusion

I have argued that climate-change models have become central contemporary environing technologies since models are key epistemic tools in climate science and the basis for much of the politics surrounding climate-change action and mitigation. Environing technology creates what we understand as environment through processes of both material alterations and cognitive framings. The results of the models are conveyed visually, and these graphs can be regarded as

environing media, which is more precise and brings the translational aspect of these graphs to the fore.

The key point in assessing climate-change temporality is to also incorporate the future in environing and to suggest that the fourth dimension of time is linked to the geographical and ideological dimensions put forward in previous scholarship. The focus has been not on how models incorporate certain Earth-system components and exclude others but rather on the temporalities climate-change models span and create. The models themselves and their input, together with the general tools of modeling, have shifting affordances in environing our future Earth.

The target year of 2100 is a paradox as it forcefully frames the climate issue temporarily, allowing for a big enough signal to show change, but at the same time being of a scale that does not easily translate into action. I suggested initially that our growing interest in history now, when our future seems to vanish, was congenial. Perhaps the paradox of the target year is congenial too, and in line with the wickedness of the climate problem itself.

Acknowledgment

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Notes

1 Martin J. S. Rudwick, *Worlds Before Adam: The Reconstruction of Geohistory in the Age of Reform* (Chicago: University of Chicago Press, 2008).

2 I am here thinking of Yuval Noah Harari, *Sapiens: A Brief History of Humankind* (London: Harvill Secker, 2014) or David Christian, *Origin Story: A Big History of Everything* (London: Allen Lane, 2018, but also Stephen Hawking, *A Brief History of Time: From the Big Bang to Black Holes* (London: Bantam Press, 1988).

3 Jenny Andersson & Eglė Rindzevičiūtė, eds., *The Struggle for the Long-Term in Transnational Science and Politics: Forging the Future* (New York: Routledge, Taylor & Francis Group, 2015). In this context see also for example Birgit Schneider, "The Future Face of the Earth: The Visual Semantics of the Future in the Climate Change Imagery of the IPCC," *Cultures of Prediction in Atmospheric and Climate Science: Epistemic and Cultural Shifts in Computer-Based Modelling and Simulation*, ed. Matthias. Heymann, Gabriele Gramelsberger, & Martin Mahony (New York: Routledge, 2017), 231–51, and Nina Wormbs, ed., *Competing Arctic Futures: Historical and Contemporary Perspectives* (Cham: Palgrave Macmillan, 2018) for a discussion on futures.

4 Paul N. Edwards, "Representing the Global Atmosphere: Computer Models, Data, and Knowledge about Climate Change," *Changing the Atmosphere: Expert Knowledge and Environmental Governance*, eds. Clark A. Miller & Paul N Edwards (Cambridge, MA: MIT Press, 2001); Mikaela Sundberg, Parameterizations as Boundary Objects on the Climate Arena," *Social Studies of Science* 37, no. 3 (1 June 2007): 473–88; https://doi.org/10.1177/0306312706075330; Martin Mahony & Mike Hulme, "Model Migrations: Mobility and Boundary Crossings in Regional Climate Prediction," *Transactions of the Institute of British Geographers* 37, no. 2 (2012): 197–211.

5 Cf. Birgit Schneider & Thomas Nocke, *Image Politics of Climate Change: Visualizations, Imaginations, Documentations* (Bielefeld: Transcript, 2014). See also a discussion on cultural techniques by the editors in this volume's introduction.
6 The empirical material for this argument overlaps with the one used in a longer book chapter, Nina Wormbs, "Model Time and Target Year: On the End of Time in IPCC Futures," *Times of History, Times of Nature: Temporalization and the Limits of Modern Knowledge*, eds. Anders Ekström & Staffan Bergwik (New York: Berghahn Books, 2022), 284–307. There I argue that we need to pay attention to what I call "model time" in the temporalities of models and their relation to our understanding of climate change, whereas I here focus on these models as environing technology.
7 They are all open access and can be found at the IPCC web: ipcc.ch. The IPCC also publishes a lot of reports and material that are not the regular assessment reports. The 2018 report on the 1.5-degree target is such an example. I do not use that material in this chapter but focus solely on the assessment reports. Different parts of the sixth report are being published as I prepare this manuscript and are not part of my empirical material.
8 Also, others judge the summaries for policy makers to have a higher circulation. See for example Lynda Walsh, *Scientists as Prophets: A Rhetorical Genealogy* (New York: Oxford University Press, 2013).
9 For a discussion on the Planetary Boundaries framework see the contribution by Sverker Sörlin, this volume.
10 See for example Mike Hulme, *Why We Disagree about Climate Change: Understanding Controversy, Inaction and Opportunity* (Cambridge: Cambridge University Press, 2009).
11 Miyase Christensen, Annika E. Nilsson, & Nina Wormbs, eds., *Media and the Politics of Arctic Climate Change: When the Ice Breaks*, 1st ed. (New York: Palgrave Macmillan, 2013); Naomi Oreskes, "The Scientific Consensus on Climate Change," *Science* 306, no. 5702 (2 December 2004): 1686.
12 See key publications like Kirsten Hastrup & Martin Skrydstrup, eds., *The Social Life of Climate Change Models : Anticipating Nature* (London: Routledge, 2013) and Matthias Heymann, Gabriele Gramelsberger, & Martin Mahony, eds., *Cultures of Prediction in Atmospheric and Climate Science: Epistemic and Cultural Shifts in Computer-Based Modelling and Simulation* (New York: Routledge, 2017). See also Shinichiro Asayama et al., "Why Setting a Climate Deadline Is Dangerous," *Nature Climate Change* 9, no. 8 (1 August 2019): 570–572, https://doi.org/10.1038/s41558-019-0543-4 who argues that it is easier to relate time to human experience, than mean temperatures and other climate metrics. There are indeed historians dealing with climate change who have taken an interest in time. See for example Dipesh Chakrabarty, "The Climate of History: Four Theses," *Critical Inquiry* 35, no. 2 (2009): 197–222.
13 IPCC First Assessment Report Overview and Policymaker Summaries and 1992 IPCC Supplement, 11.
14 Edwards, *A Vast Machine*; Spencer R. Weart, *The Discovery of Global Warming* (Cambridge, MA: Harvard University Press, 2008). See also the contribution by Christoph Rosol, this volume.
15 This builds on my analysis of the summaries for policy makers of all five reports.
16 Anne Pasek, "Mediating Climate, Mediating Scale," *Humanities* 8, no. 4 (2019), DOI: 10.3390/h8040159.
17 Lynda Walsh, "The Visual Rhetoric of Climate Change," *WIREs Climate Change* 6, no. 4 (1 July 2015): 361–368, https://doi.org/10.1002/wcc.342, page 362.
18 James S. Risbey, "The New Climate Discourse: Alarmist or Alarming?" *Global Environmental Change* 18, no. 1 (1 February 2008): 26–37, https://doi.org/10.1016/j.gloenvcha.2007.06.003. Quote page 33.
19 Risbey, 33.
20 Pasek.

21 Pasek. The acquired position also meant that climate skeptics chose to attack it.
22 Helge Jordheim, "Introduction: Multiple Times and the Work of Synchronization," *History and Theory* 53, no. 4 (1 December 2014): 498–518, https://doi.org/10.1111/hith.10728. For an elaboration see Wormbs (2022).
23 Apparently, 15 years is a timeframe that is most suitable for humans to overview. See Sabine Pahl et al., "Perceptions of Time in Relation to Climate Change," *WIREs Climate Change* 5, no. 3 (1 May 2014): 375–388, https://doi.org/10.1002/wcc.272.
24 Bård Lahn, "A History of the Global Carbon Budget," *WIREs Climate Change* 11, no. 3 (1 May 2020): e636, https://doi.org/10.1002/wcc.636. It has been criticized by, for example, Asayama et al.
25 Sverker Sörlin & Paul Warde, eds., *Nature's End: History and the Environment* (Houndmills, Basingstoke: Palgrave Macmillan, 2009); Sverker Sörlin, "Environment," *Companion to Environmental Studies*, eds. Noel Castree, Mike Hulme, & James D. Proctor (New York: Routledge, 2018).
26 The concept environing technology was first coined by Sverker Sörlin in 2011 and subsequently discussed at the Division of History at KTH, Royal Institute of Technology over several years. Key publications discussing the term are Sverker Sörlin & Nina Wormbs, "Environing Technologies: A Theory of Making Environment," *History and Technology* 34, no. 2 (3 April 2018): 101–125, https://doi.org/10.1080/07341512.2018.1548066; Johan Gärdebo, *Environing Technology: Swedish Satellite Remote Sensing in the Making of Environment 1969–2001* (Stockholm: KTH Royal Institute of Technology, 2019); Sabine Höhler & Nina Wormbs, "Remote Sensing : Digital Data at a Distance," *Methodological Challenges in Nature-Culture and Environmental History Research*, ed. Stephanie Rutherford Jocelyn Thorpe L. Anders Sandberg (London, New York: Routledge, 2017), 272–283.
27 Gabrielle Hecht, "Inside-Out Earth," manuscript in review, and session "The Anthropocene as a provocation to the history of technology – a performative debate" at the annual meeting with the Society for the history of technology, Milan, October 26th, 2019.
28 Nina Wormbs, & Johan Gärdebo, "The Distant Gardener: Remote Sensing of the Planetary Potager," *Gardens and Human Agency in the Anthropocene*, eds. Maria Paula Diogo et al. (London: Routledge, 2019), 124–142.
29 Stig Hjarvard, "The Mediatization of Society: A Theory of the Media as Agents of Social and Cultural Change," *Nordicom Review* 29 (2008): 105–134.
30 Susan J. Douglas, "Some Thoughts on the Question 'How Do New Things Happen?'" *Technology and Culture* 51, no. 2 (2010): 293–304.
31 Ian Hutchby, "Technologies, Texts and Affordances," *Sociology* 35, no. 2 (2001): 441–456. Technology as a concept is also elusive and an object of debate. For a recent reflection see John Agar, "What Is Technology?" *Annals of Science*, Palgrave studies in the history of science and technology, 77, no. 3 (2020): 377–382. On a more general level, Thomas Misa has suggested that looking at a meso-level for analysis can avoid black boxing while at the same time allow for enough but not too much case-specific detail. Thomas J. Misa, "Retrieving Sociotechnical Change from Technological Determinism," *Does Technology Drive History? The Dilemma of Technological Determinism*, eds. Merritt Roe Smith & Leo Marx, vol. 1994 (Cambridge, MA: MIT Press, n.d.), 101–114.
32 John Durham Peters, *The Marvelous Clouds: Toward a Philosophy of Elemental Media* (Chicago: The University of Chicago Press, 2015) and this volume.
33 For a discussion on environment as a temporal category see also Sverker Sörlin, "Environmental Times: Synchronizing Human-Earth Temporalities from Annales to Anthropocene, 1920s–2020s," *Times of History, Times of Nature: Temporalization and the Limits of Modern Knowledge*, eds. Anders Ekström & Staffan Bergwik (New York: Berghahn Books, 2022), 64–101.

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